

- Analysis and Calculus on $\mathbb{R}^n$.

- What is analysis?

Bound things like $|x_n - a| < \varepsilon$ and so fourth.

- What is calculus?

Arithmetics ($+$, $-$, $\times$, $\div$) plus two operations

Differentiation: d

Integration: $\int$

- What I will go over rather quickly:

$\rightarrow$ sup, inf, lim, open/closed sets, compact sets.

You will see these again in Math Method in the fall.

- What I will spend more time on:

Differential calculus

- A bit time on: Riemann Integration.

- Bounds and completeness of $\mathbb{R}$.

"Completeness" = "No gaps". Real line has no gaps.

To talk about completeness, we introduce the language of bounds.

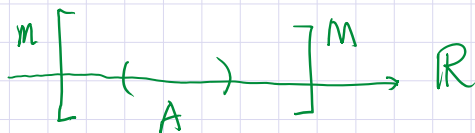
M is an upper bound of $A \subseteq \mathbb{R}$ if

$$x \leq M \text{ for all } x \in A$$

m is a lower bound of $A \subseteq \mathbb{R}$ if

$$x \geq m \text{ for all } x \in A$$

$$x \geq m \text{ for all } x \in A$$



Note bounds $\neq$ maximum/minimum. Precisely ^{Maximum}_{minimum} are bounds

$$\underline{a^*} \in A \text{ max if } x \leq a^* \forall x \in A$$

$$\underline{a_*} \in A \text{ min if } x \geq a_* \forall x \in A$$

- Supremum and Infimum:

Let $A \subseteq \mathbb{R}$, $A \neq \emptyset$ (empty set)

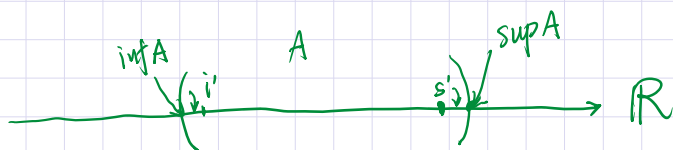
$$s = \sup A \text{ if } \begin{array}{l} \textcircled{1} x \leq s \text{ for all } x \in A \\ \textcircled{2} s \text{ is the least upper bound} \end{array}$$

For every $s' < s$ there exists $x_0 \in A$ such that $s' < x \leq s$

$$\bar{i} = \inf A \text{ if } \begin{array}{l} \textcircled{1} x \geq \bar{i} \text{ for all } x \in A \\ \textcircled{2} \bar{i} \text{ is the greatest lower bound:} \end{array}$$

For every $i' > \bar{i}$, there exists $\tilde{x} \in A$ such that $i' > \tilde{x} \geq \bar{i}$.

When in doubt, draw a picture:



- ϵ -Language for sup/inf : $A \neq \emptyset$

If $s = \sup A$, then for every $\epsilon > 0$:

There exists some $x_\epsilon \in A$ such that $\overbrace{s-\epsilon}^{s'} < x_\epsilon \leq s$

If $\bar{i} = \inf A$ then for every $\epsilon > 0$:

There exists some $x_\epsilon \in A$ such that $\overbrace{\bar{i}+\epsilon}^{i'} > x_\epsilon \geq \bar{i}$

There exists some $x_\varepsilon \in A$ such that $\tilde{i} + \varepsilon > x_\varepsilon \geq i$

- Simple Example: $A = [0, 1)$

$$\sup A = 1 \notin A$$

$$\inf A = 0 \in A$$

then $0 = \min A$ and A has no max.

- Completeness means that every non-empty subset that is bounded above has a unique sup. $\left. \begin{array}{l} \text{that is} \\ \updownarrow \\ \text{that is} \end{array} \right\} \begin{array}{l} \text{bounded below} \\ \text{has a unique inf.} \end{array} \left. \vphantom{\begin{array}{l} \text{that is} \\ \updownarrow \\ \text{that is} \end{array}} \right\} \text{Completeness Axiom.}$

- $\mathbb{Q}$ is NOT complete.

Take: $A = \{x \in \mathbb{Q} : x^2 < 2\}$ bounded No sup/inf in $\mathbb{Q}$.

- Completeness means that $A \neq \emptyset$ bounded in $\mathbb{R}$

$\downarrow$
 $\sup A / \inf A$ exists in $\mathbb{R}$

- Archimedean Property:

1. The natural numbers are not bounded above in $\mathbb{R}$

2. For every $\varepsilon > 0$, there exists $n_\varepsilon \in \mathbb{N}$ such that

$$0 < \frac{1}{n_\varepsilon} < \varepsilon$$

Proof: 1 By contradiction: Suppose $\mathbb{N}$ is bounded above in $\mathbb{R}$

By completeness of $\mathbb{R}$ we have $s \in \mathbb{R}$

$$s = \sup \mathbb{N}.$$

(taking $\varepsilon = 1$)

Since $s - 1 < s$ by the ε -language / defn of

Supremum: $s - 1 \leq n \leq s$

Supremum: $S-1 < n \leq S$

Then $n+1 > S$ but $n+1 \in \mathbb{N}$

contradicting that S is an upper bound of $\mathbb{N}$.

2. Let $\varepsilon > 0$, Since $\mathbb{N}$ is NOT bounded above, there exists $n_\varepsilon \in \mathbb{N}$ such that

$$n_\varepsilon > \frac{1}{\varepsilon} \Leftrightarrow 0 < \frac{1}{n_\varepsilon} < \varepsilon$$

□

• Fact: Equivalent definitions of completeness:

Every Cauchy sequence converges to a limit in $\mathbb{R}$

[A sequence $\{a_n\}_{n=1}^\infty$ is Cauchy if for any $\varepsilon > 0$ there exists $N_\varepsilon \in \mathbb{N} : \forall m, n > N_\varepsilon$
 $|a_m - a_n| < \varepsilon$]

- Norms on $\mathbb{R}^n = \{x = (x_1, \dots, x_n)^T : x_i \in \mathbb{R}\}$

• Dot product: $x \cdot y = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$

• Euclidean Norm (2-Norm):

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + \dots + x_n^2}$$

• Euclidean Distance: $d_2(x, y) = \|x - y\|$

Triangle Inequality of $\|\cdot\|$ gives $d_2(x, z) \leq d_2(x, y) + d_2(y, z)$.

→ General Principle in Analysis:

Inner Product $\rightsquigarrow$ Norm $\rightsquigarrow$ Distance / Metric

$\langle \cdot, \cdot \rangle$ $\| \cdot \|$ $d(x, y)$

$\mathbb{R}^n$ Dot product Euclidean Euclidean Distance

$\langle \cdot, \cdot \rangle$
 $\mathbb{R}^n$: Dot product

$\|\cdot\|$
Euclidean Norm

usually,
Euclidean Distance

- Convergence in $\mathbb{R}^n$:

A sequence $\{x_k\}_{k=1}^{\infty} \subseteq \mathbb{R}^n$ converges to $a \in \mathbb{R}^n$ if

$$\|x_k - a\| \rightarrow 0 \quad \text{as } k \rightarrow \infty$$

Equivalently: (ϵ - N Language)

$\{x_k\}$ converges to a : if for every $\epsilon > 0$, there exists
 $N_{\epsilon} \in \mathbb{N}$: For all $k > N_{\epsilon}$
 $\|x_k - a\| < \epsilon$.

We write $\lim_{k \rightarrow \infty} x_k = a$ or $x_k \rightarrow a$ ($k \rightarrow \infty$).

If $x_k = (x_k^1, x_k^2, \dots, x_k^n)^T$ and $a = (a^1, a^2, \dots, a^n)$

Then $x_k \rightarrow a \iff x_k^i \rightarrow a^i$ for all i .

Proof uses: $\forall 1 \leq i \leq n$:

$$|x_k^i - a^i| \leq \|x_k - a\| \leq \sqrt{n} \max_i |x_k^i - a^i|$$

□

Coming Next: Open / Closed Sets. Compact Sets (Brief).

Continuity
Differentiation.